

App Dev Project Report

Trekking Management Application

1. Student Details

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About Me: I am 24 batch BS Data Science student with an interest in full-stack web development, backend engineering, and machine learning. I enjoy building secure, scalable, and user-friendly applications.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

To develop a role-based web application for managing trekking activities where administrators manage treks, staff, users, and bookings while trekkers can browse, book, and track their trekking history.

Approach:

The project was developed using Flask with a modular Blueprint architecture. SQLAlchemy and SQLite manage persistent storage, Flask-Login provides authentication, Bootstrap 5 ensures a responsive interface, and Jinja2 renders dynamic pages.

3. AI/LLM Declaration

AI Tool Used: ChatGPT (OpenAI GPT-5)

Estimated AI Usage: 15–20%

AI assistance was limited to frontend styling, database designing. All implementation, testing, integration, validation, and architectural decisions were completed by me.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend framework
Flask-SQLAlchemy	ORM
SQLite	Database
Flask-Migrate	Database migrations
Flask-Login	Authentication
Jinja2	Template engine
Bootstrap 5	Responsive UI
Werkzeug	Password hashing
HTML5/CSS3	Frontend

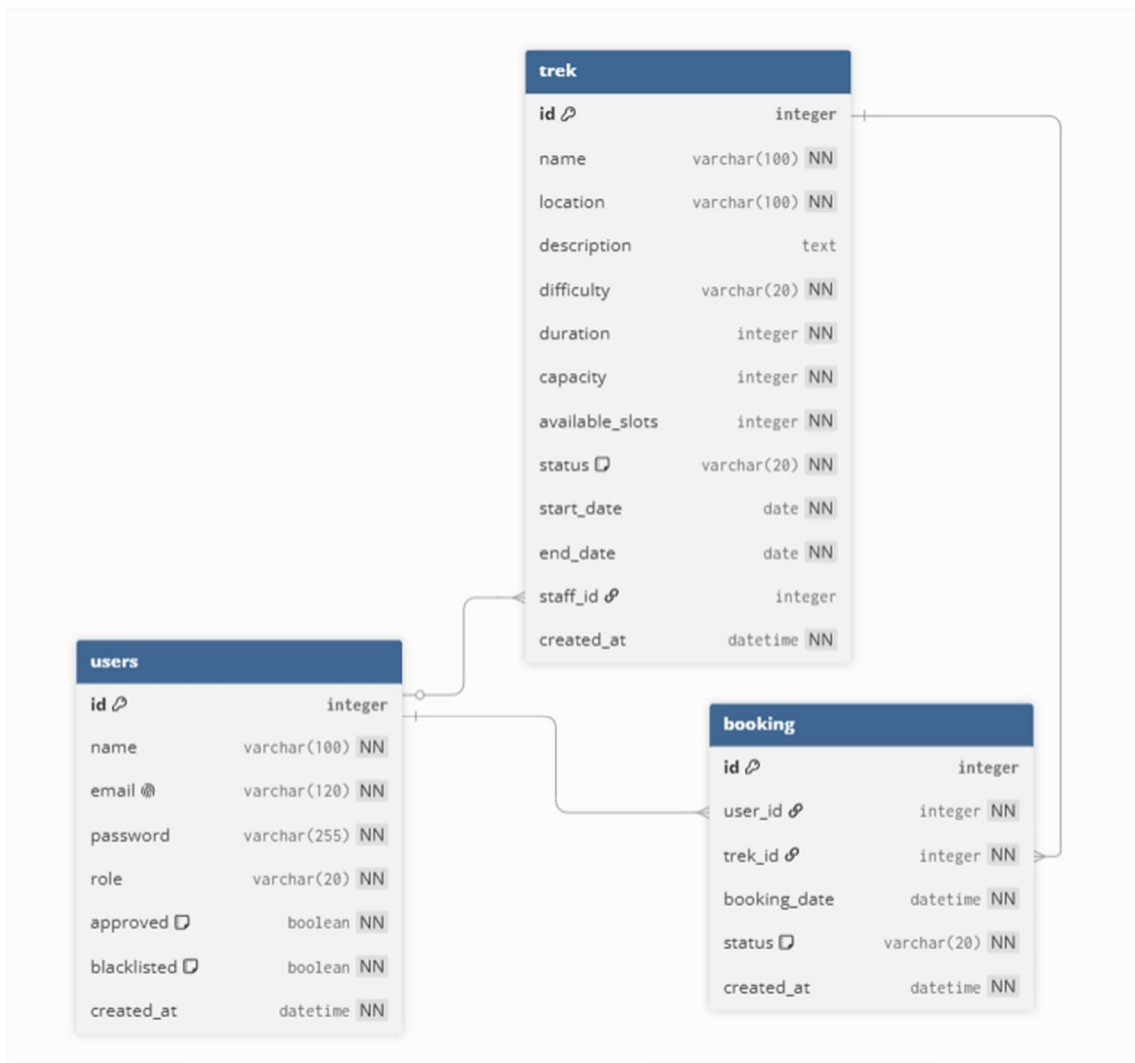
5. Database Schema / ER Diagram

Tables:

- User
- Trek
- Booking

Relationships:

- One Staff → Many Treks
- One User → Many Bookings
- One Trek → Many Bookings



6. API Resource Endpoints

Endpoint	Method	Description
/login	GET/POST	Authenticate user
/register	GET/POST	Register staff/user
/admin/treks	GET	Manage treks
/user/treks/<id>/book	POST	Book trek
/user/bookings	GET	View bookings
/staff/treks/<id>	GET/POST	Manage assigned trek
/profile	GET/POST	Update profile

7. Architecture and Features

Architecture Overview:

- app.py (entry point)
- models/ (SQLAlchemy models)
- routes/ (Blueprint controllers)
- templates/ (Jinja2 views)
- static/ (assets)
- utils/ (decorators/helpers)

Implemented Features:

Role-based authentication
Admin dashboard
Staff dashboard
User dashboard
Trek booking and cancellation
Booking history
Trek status tracking
Responsive Bootstrap UI
Server-side validation

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1_KJ_6Gk1xMkFuSYL1eISkWOXcAjxLe25/view?usp=sharing